

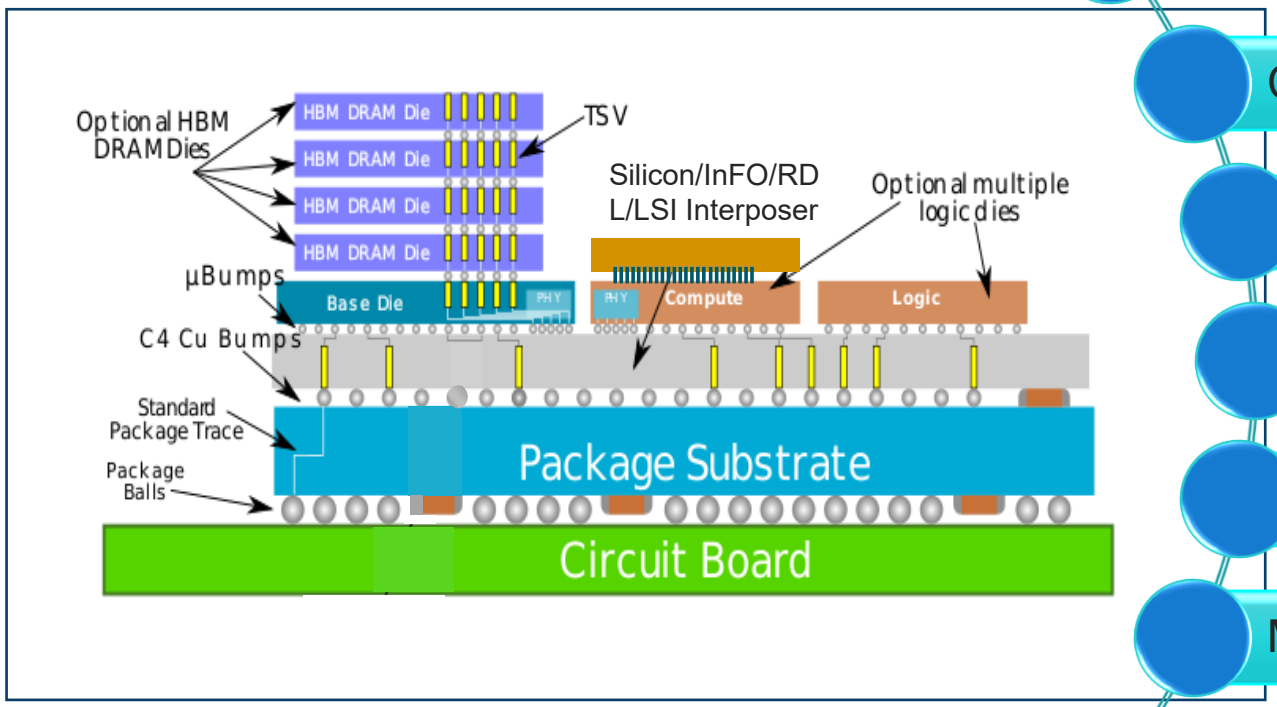
Cadence Multi-Die/Multi-Chiplet Solution Overview

Unified solution for efficient design

Chiplet Summit 2026

cadence®

3D-IC Full-flow Requirements for Complex Multi-Chiplet Designs



- System Partitioning, Planning, and Prototyping
- Chiplet Implementation and System PPA
- Analog and Package Co-design
- Thermal and Power (Early Analysis and Signoff)
- Multi-Die Extraction and Timing Signoff
- Multi-Die Physical Verification
- System/Package-Level Extraction SI/PI Analysis

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Integrity 3D-IC Platform: A Complete Solution for Multi-Chiplet Design

Unified Platform

- Efficient design aggregation
- Full-flow on unified database

Leading Implementation Technology with 3D Upgrades

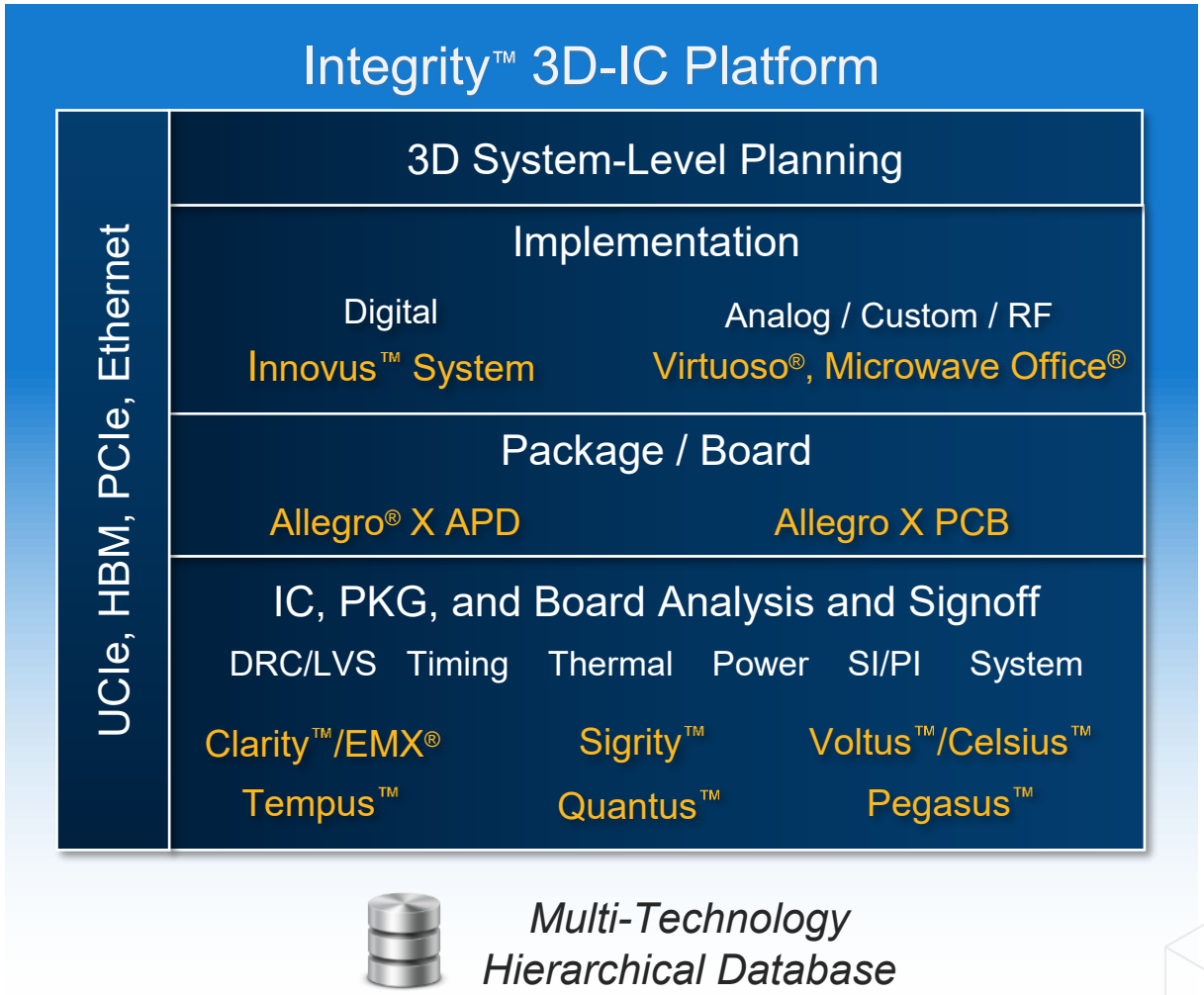
- Innovus™/Virtuoso®: Digital/analog chip implementation
- Allegro® Package Designer: Package implementation

Integrated Analysis – Chip + Package

- IR, thermal, and warpage analysis
- Signal integrity, extraction, STA, system LVS/DRC

Foundry Qualified

- Full support for 3Dblox (IEEE P3537)
- TSMC - Certified flows for all 3DFabric offerings
- Qualified flows at all other foundries



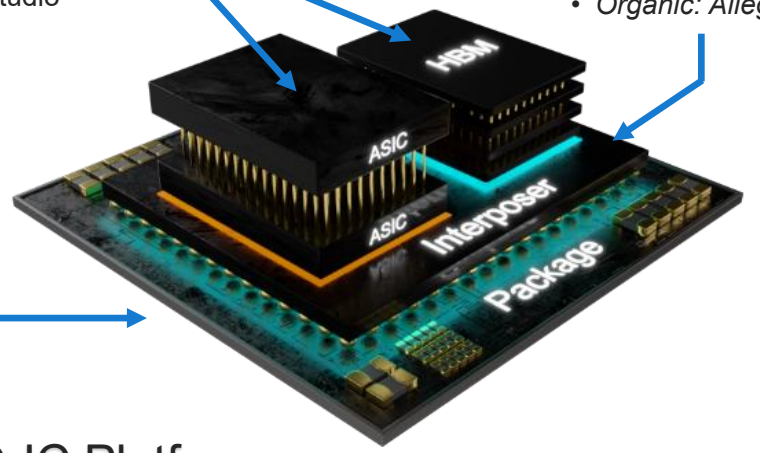
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Technology for Designing and Analyzing the Whole Stack

CHIP STACKING
Implemented in Innovus™ System or Virtuoso® Studio

INTERPOSER
Off-chip implementation and analysis
- *Silicon: Innovus or Virtuoso Studio*
- *Organic: Allegro X APD*

PACKAGE IMPLEMENTATION
Package routing with Allegro® X APD



Integrity™ 3D-IC Platform

3D System-Level Planning	Integrity™ System Planner
3D Implementation	Innovus™ System, Virtuoso® Studio, Allegro® X APD
Package Implementation	Allegro X APD
3D Analysis and Signoff	**Die:** Pegasus™, Quantus™, Tempus™, Voltus™ solutions **Interposer/Package:** Sigrity™, Clarity™, Celsius™ solutions
Die-to-Die IP	UCIe, HBM, PCIe, Ethernet

Integrity™ 3D-IC Platform

UCIe, HBM, PCIe, Ethernet

3D System Level Planning

Implementation

Digital
Innovus™ System

Analog / Custom / RF
Virtuoso®, Microwave Office

Package / Board

Allegro® X APD

Allegro X PCB

IC, PKG, and Board Analysis and Signoff

DRC/LVS Timing Thermal Power SI/PI System

Clarity™/EMX® Sigrity™ Voltus™/Celsius™
Tempus™ Quantus™ Pegasus™

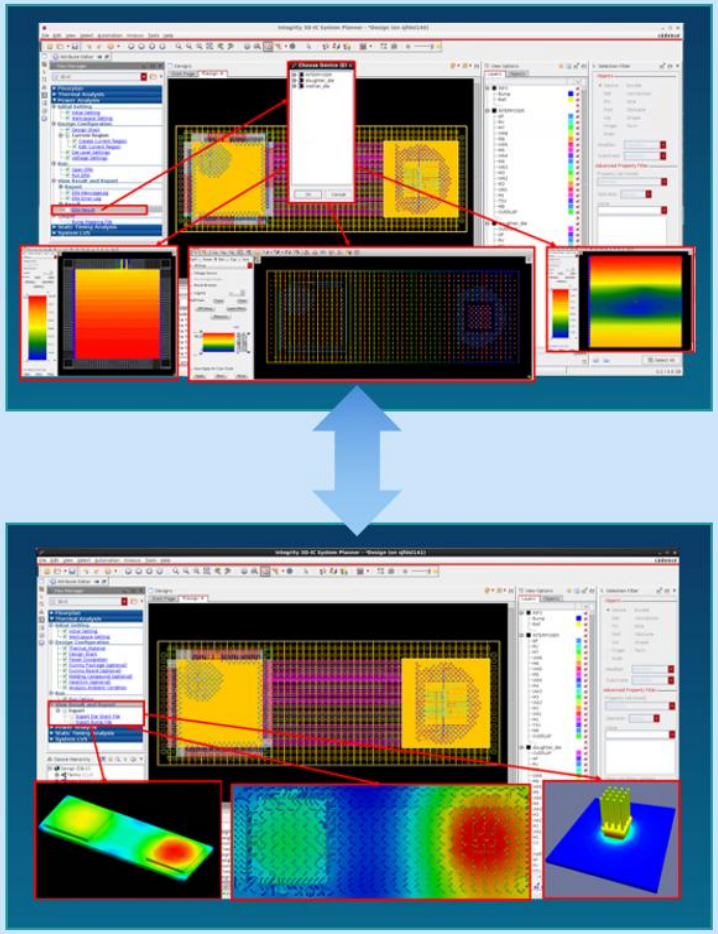
Multi-Technology Hierarchical Database

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System-Level Analysis/Signoff Flow Integration Status

Voltus™/Celsius™

Early Rail (Power and Thermal) Analysis



System-Level Power and Thermal Analysis

Integrity™ Flow Manager

Guided Flow

2 Implementation

3 PDN Analysis

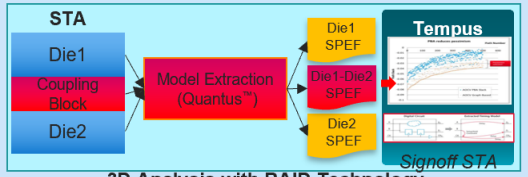
4 Thermal Analysis

5 Static Timing Analysis

6 System LVS

7 SI/PI Analysis

Tempus™ *Timing Analysis*



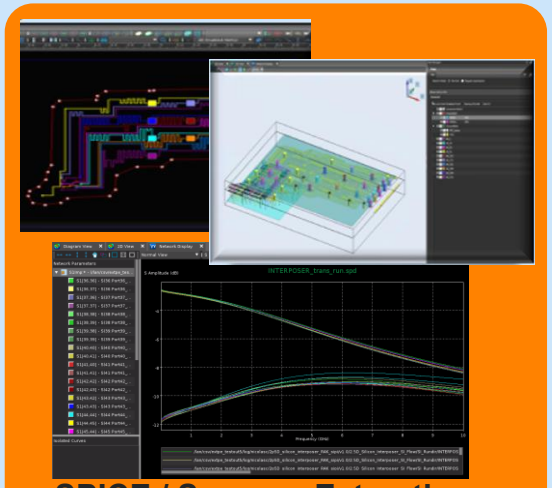
3D Analysis with RAID Technology

Pegasus™ *Physical Verification (DRC/LVS)*

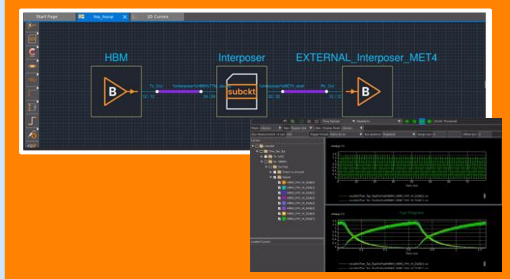


Inter-Die Connectivity Checking

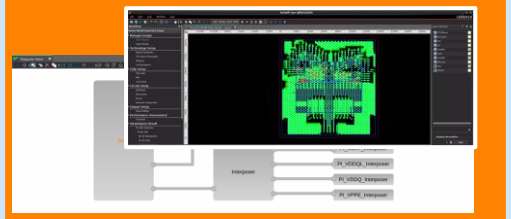
Sigrity™ Tools SI/PI Analysis Flows



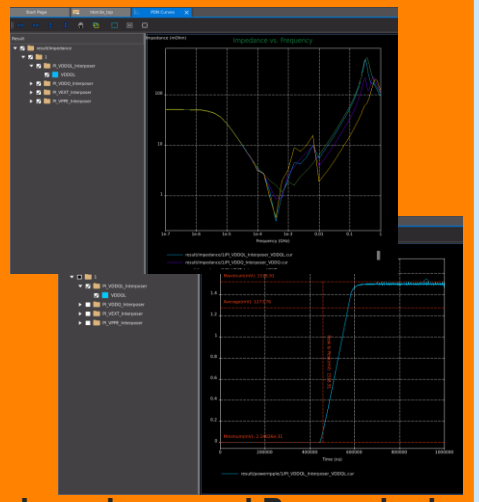
SPICE / S-param Extraction



SI - Eye-diagram Analysis



SPICE / S-param Extraction



PI - Impedance and Power ripple

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RAKs

Basic Silicon Interposer 2.5D, 3D Stacking, Silicon-Bridge

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RAKs and Documentation Update http://support.cadence.com/3dic

- 2.5D – Advanced Packaging RAKs
  - 2.5D Silicon Interposer System Codesign Using Integrity 3D-IC Platform
  - 2.5D Package Codesign Using Integrity 3D-IC Platform
  - 2.5D RDL Interposer System Codesign Using Integrity 3D-IC Platform and Allegro PD Plus

- 3D – Silicon stacking RAKs
  - 3D RISC-V Codesign and Analysis Using Integrity 3D-IC Platform
  - 3D RISC-V ERA PSDL Flow Using Integrity 3D-IC Platform
  - 3D RISC-V ERA DEF Flow Using Integrity 3D-IC Platform
  - 3D RISC-V Digital Quantus Flow Using Integrity 3D-IC Platform
  - 3D RISC-V Thermal Flow Using Integrity 3D-IC Platform
  - 3D RISC-V STA Flow Using Integrity 3D-IC Platform
  - 3D RISC-V System LVS Flow Using Integrity 3D-IC Platform



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Integrity 3D-IC Rapid Adoption Kit - Example

Available on Cadence Online Support

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Designing with the Integrity 3D-IC Platform

Rapid Adoption Kit (RAK)

Product Version: Integrity 3D-IC 21.12
December 2021

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Customer Examples

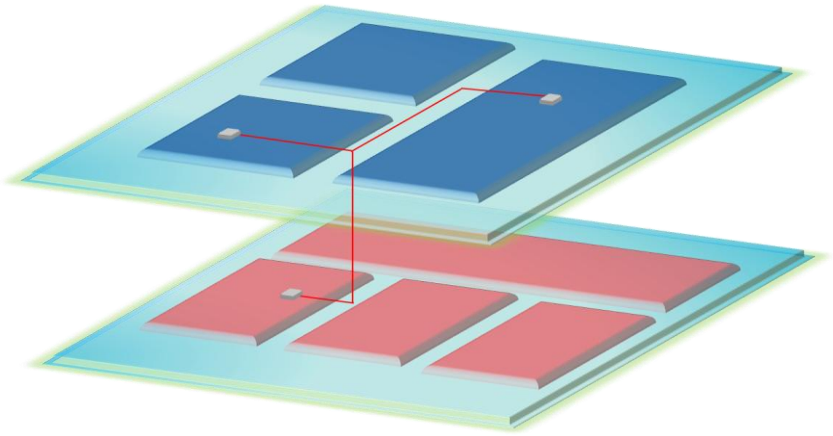
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Integrity 3D-IC Customer Proliferation

Customer	Type	Integrity 3D-IC Activity/Benefits
CUSTOMER A	2.5D Tapeout	AI accelerator (CoWoS-S). System Planning, SI Interposer Routing with SI verification. Increased Productivity and TAT.
CUSTOMER B	3D TestChip 3D Prototyping	Form factor limited design 50% implementation resources, 20% area, 20% wirelength savings Prototyping and exploration using SAI for large datacenter design.
CUSTOMER C	2.5D Tape-Outs 3.5D SoIC Tapeout	3.3X Interposer tapeout. High shield interposer routing with hierarchy. 70% cycle time reduction 3D Heterogenous stack db. Large capacity bump handling. Efficient system planning.
CUSTOMER D	3D Core Folding HBM-Pkg ref. flow	Project 1: High-performance core. 21% wirelength, 43% area improvement. Project 2: 30% Productivity improvement with integrated thermal/SI/PI flows.
CUSTOMER E	3D SoIC Tapeout	Efficient stack planning. 7X improvement in 3DIR solver, 10X GUI improvement.
CUSTOMER F	3.5D SoIC virtual TO	3D stacked die data management, bump and TSV optimization, chiplet PDV planning.
CUSTOMER G	SoW Tapeout	Full system-on-a-wafer inFO Design. First time flow and tapeout for Dojo.
CUSTOMER H	3D Tapeout	Efficient management of multiple image sensor chips in a unified environment
CUSTOMER I	3D Core Folding	30% power reduction i(D2D). Up to 0.85X area scaling.
CUSTOMER J	3D Photonics	High-capacity data management. 60K-uBumps, 3K -TSVs, 10K - C4 bumps.
CUSTOMER K	2.5D Tapeout 3.5D SoIC Tapeout	2.5D tapeout data management. High shield interposer routing. Efficient Bump assignment with integrated Innovus™, System LVS/DRC.
CUSTOMER L	3D TestChip	38% better performance, 13% power savings, unified thermal env. 3-tier 3D-IC.
CUSTOMER M	3D Chiplet flow	Seamless 3Dblox based flow for 2.5D and 3DChiplet stacking.
CUSTOMER N	3D Exploration	2.5D design tapeout with full flow. Integrated platform features used for interposer routing and system planning
CUSTOMER O	3D Testchip	2 tapeouts – Wafer-on-Wafer and Die-on-Wafer. 3D Full digital flow exercised.
CUSTOMER P	2.5D Tapeout	2.5D chiplet integration - system connection with interposer and package.

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3D Heterogeneous Prototyping and Design Flow with Chiplets



Design Details		
Top	3nm	Logic and MBIST
Bottom	5nm	Logic and MEM array
Interface	Face-2-Face with Hybrid Bonding	**Synchronous 3D Clock Mesh**

Requirements		Cadence Solution
3D-IC Prototyping	Die stacking and IR impacts	Early Rail Analysis
	Die stacking and thermal impacts	Early Thermal Analysis
	TSV pattern and IC/PKG/PCB design rule	Early Power Noise Analysis
3DIC Planning	Large scale 3D design partitioning	3D-EFS
	BUMP/TSV planning (especially, for DIEs with different shrink-factors)	I3D TSV/BUMP planning
	BUMP bundle requirement	Hybrid bundle flow
	BUMP/TSV number/clustering design	TSV/BUMP analysis and congestion/timing driven concurrent placement
	BUMP alignment and connectivity check	systemLVS
3DBlox Support		3DBlox flow support and ref scripts
Implementation	cross-die placement	Global MMMC and concurrent placemen
	Hign speed inter-DIE timing	3D clock mesh flow
Signoff	3D EM-IR	Voltus 3D EM-IR flow and massive TSV array model
	3D RC extraction	Quantus IDX flow
	3D STA	Tempus 3D view reduction and STA flow
	Metal density extraction	Thermal analysis flow
	Massive TSV array modeling/Multi tech files	Power Integrity flow
	Multi-tech files	Signal Integrity flow

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Rapid Prototyping Flow in Integrity 3D-IC Platform

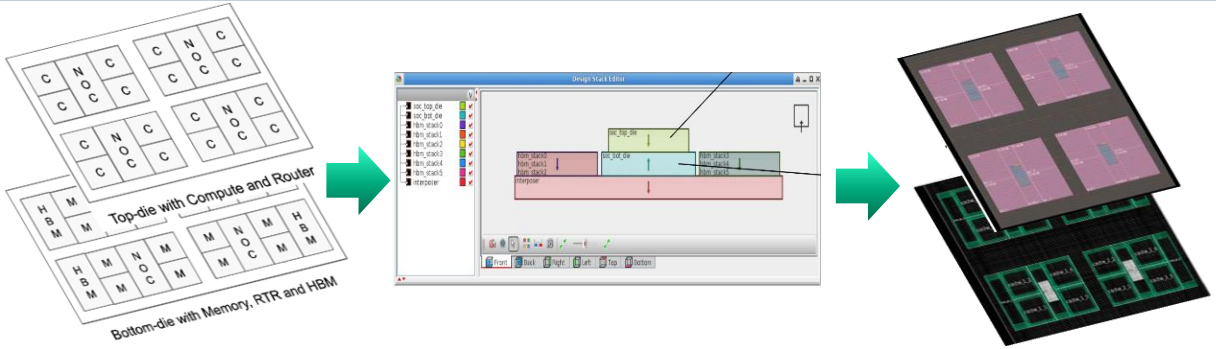
Deployed by customers to plan the next generation of 3D-IC stacked die design

Requirements

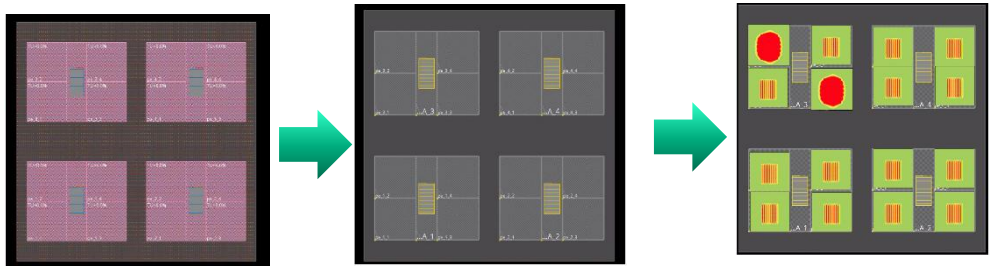
- Prototyping for early 3D exploration and pathfinding work
- Quick feedback to design teams during the planning phase
- Automation needed to construct and scale various 3D-IC prototypes on a common architecture using simple text inputs
- Meaningful prototype methodology required with early IR and thermal studies

Solution with Cadence 3D-IC technology

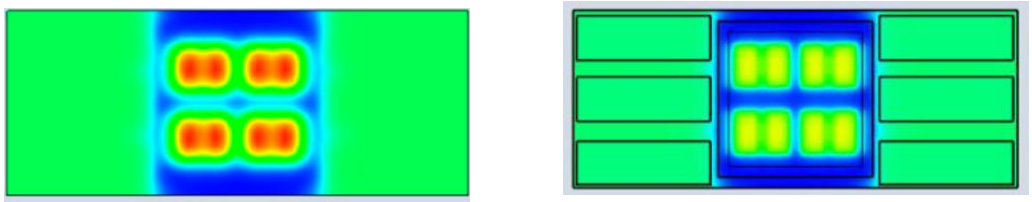
- Prototype construction and TSV/Bump planning done with Integrity™ System Planning.
- Highly scalable SAI framework from Innovus™ converts architectural specs into a technology-mapped black-box netlist
- Adaptable to real use cases with a simple change to the Gsheet input
- Connectivity and structural checks performed to verify 2D and 3D connectivity
- With bump connectivity established, early thermal and IR flow prototyping was completed



Easy translation from planning spreadsheet to Prototype using **Integrity 3D-IC** and **Innovus™ SAI**



Early Rail Analysis with PG Implementation and Hierarchical XPGV flow with **Voltus™**



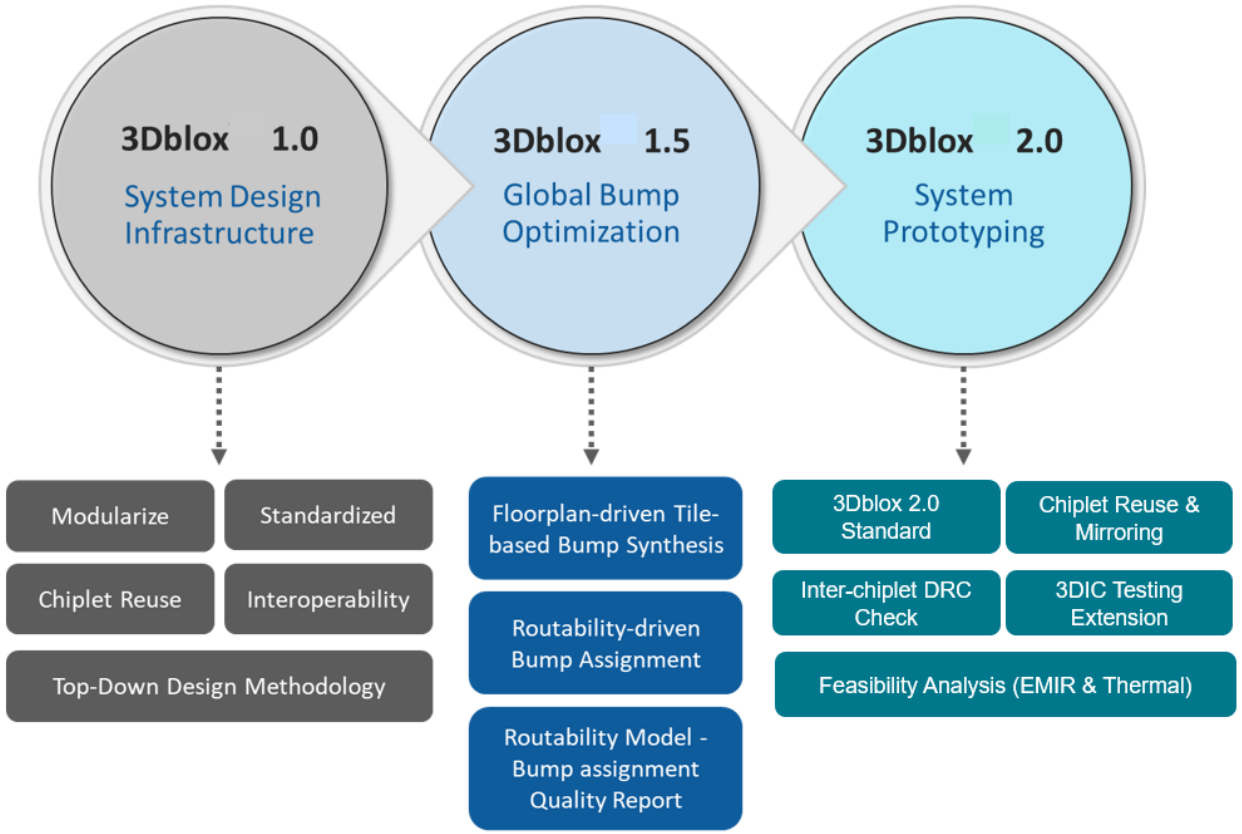
Rapid Prototyping Thermal analysis result with **Celsius™**

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Integrity 3D-IC + 3Dblox (standard)

What is 3Dblox™?

- Top-down methodology aiming at modularization and chiplet reuse
- Language constructs for configuration definition – chiplet definition, instantiation, interface, connection
- Connectivity assurance – logical assertion and physical assertion
- Different levels of abstractions – blackbox, LEF/DEF, GDS stages
- Early exploration and analysis – PDN, thermal, inter-die connectivity and DRC checking

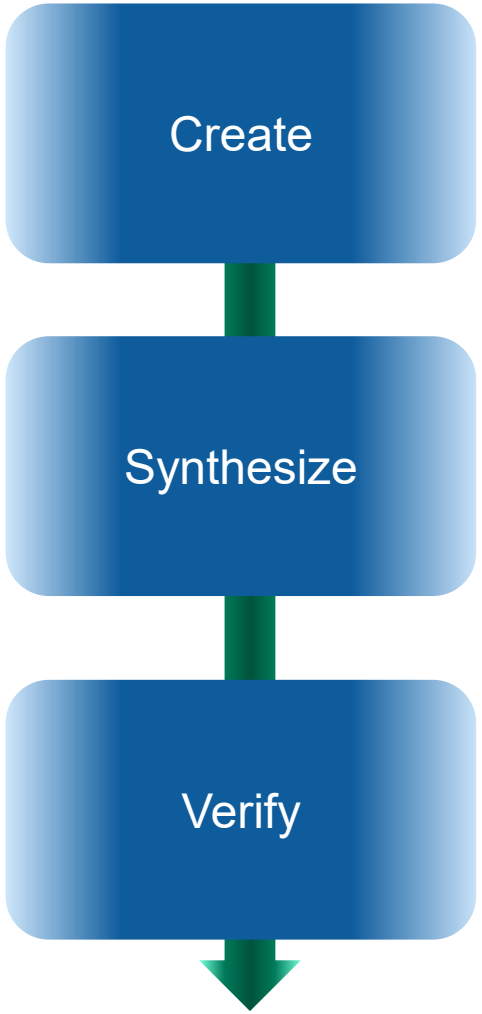


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3Dblox Modularization Flow

Simple flow steps

- Create Chiplet Definition
- Create Chiplet Instance
- Read Verilog/UPF
- Create 3D Stacking and Check
- Synthesize 3D Netlist (with LEC)
- Early DRC Check (Signoff Deck)
- Early Interface LVS
- Early PDN
- Early Thermal



Simple Chiplet definition Templates

3dbv

```
ChipletDef:
bottom_die:
    type: die
    design_area: [320.0, 320.0]
    thickness: 100.0
    seal_ring_width: [0, 0, 0, 0]
    scribe_line_remaining_width: [0, 0, 0, 0]
    shrink: 1.0
    tsv: true
    regions:
      r1:
        layer: Metal9
        side: front
        coords:
        - [0.0, 0.0]
        - [320.0, 0.0]
        - [320.0, 320.0]
        - [0.0, 320.0]
        bmap:
Collateral/bottom_die/design/bottom_die_r1.bmap
      r2:
        layer: MB
        side: back
        coords:
        - [0.0, 0.0]
        - [320.0, 0.0]
        - [320.0, 320.0]
        - [0.0, 320.0]
        bmap:
Collateral/bottom_die/design/bottom_die_r2.bmap
    external:
      APR_tech_file:
[Collateral/bottom_die/tech/*.lef]
      LEF_file:
[Collateral/bottom_die/lib/*.lef]
```

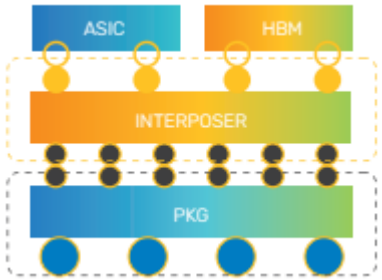
3dbx

```
ChipletInst:
  ibottom:
    master: bottom_die
    external:
      verilog_file:
      -
Collateral/bottom_die/design/bottom_die.v
      def_file:
      -
Collateral/bottom_die/design/bottom_die.def
      upf_file:
      -
Collateral/bottom_die/design/bottom_die.upf
      sdc_file: []
      gds_file: []
  itop:
    master: top_die
    external:
      verilog_file:
      - Collateral/top_die/design/top_die.v
      def_file:
      - Collateral/top_die/design/top_die.def
      upf_file:
      - Collateral/top_die/design/top_die.upf
      sdc_file: []
      gds_file: []
Stack:
  itop:
    loc: [0.0, 0.0]
    face: down
    orient: R0
    z: 100.0
    root: false
  ibottom:
    loc: [0.0, 0.0]
    face: up
    orient: R0
    z: 0.0
    root: true
```


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3Dblox System Design Infrastructure

3D-IC Structure



Bump Module | Via Module

Cap. Module | Die Module

Interposer Module

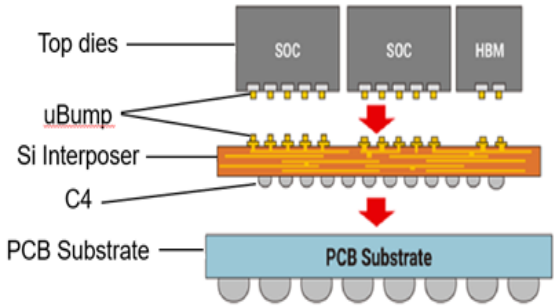
3Dblox™ Language Structure (YAML-based)

3dbv	3dbx
▪ Chiplet type ▪ Chiplet size ▪ Bump locations ▪ Bump regions ▪ Support multi-files in one design	▪ Design information ▪ Chiplet instantiation ▪ Stacking configuration ▪ Connectivity declaration ▪ Path assertion ▪ One design one 3dbx

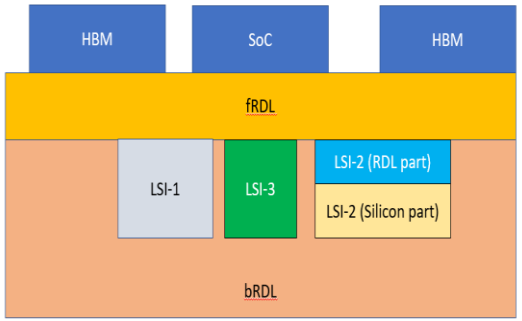
Top-Down Design Methodology

- Chiplet Definition
- Chiplet Instantiation
- Connectivity Declaration
- Physical Assertions
- Path Assertions

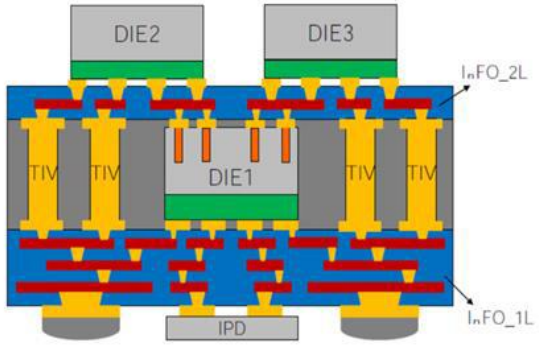
CoWoS-S



CoWoS-L

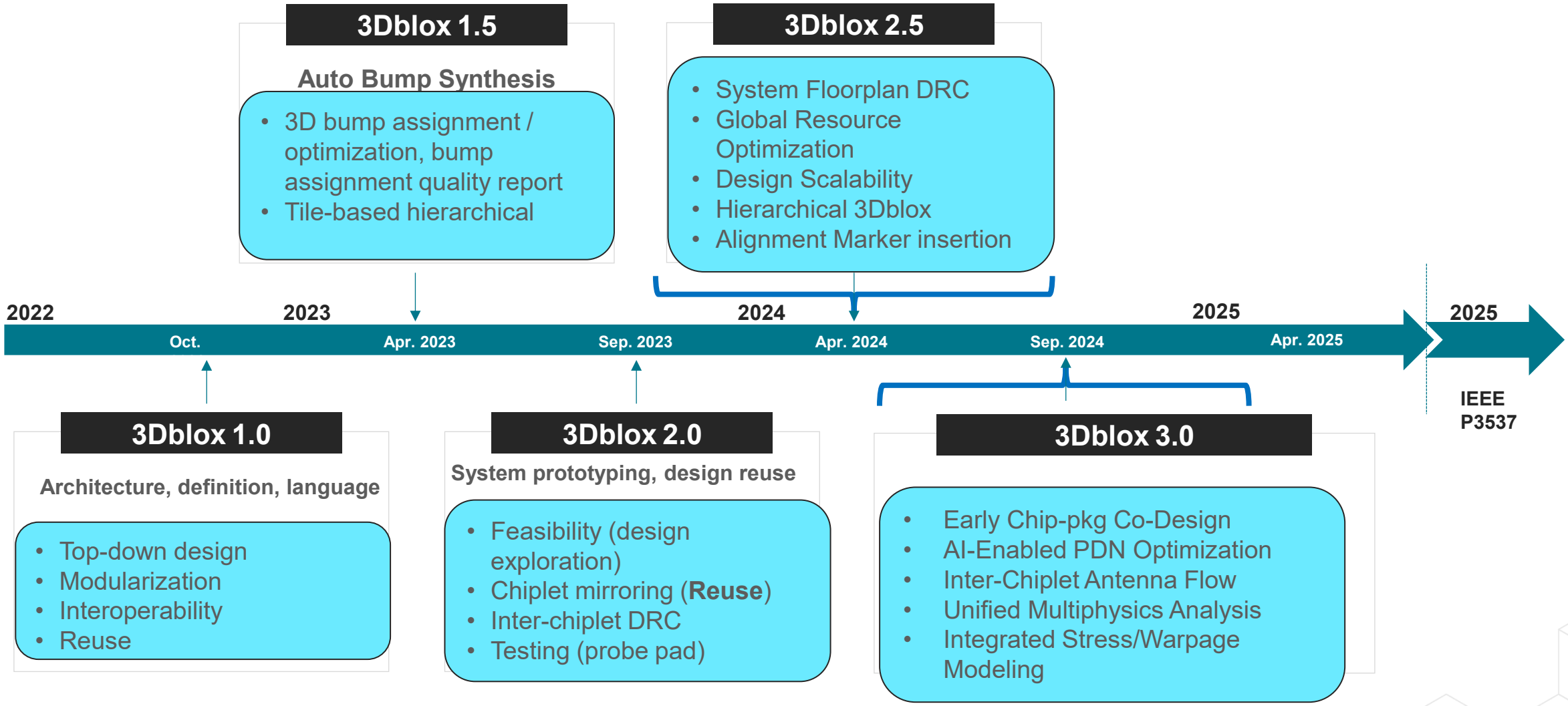


InFO-3D



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3Dblox Feature Development Timeline



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3Dblox-based Design Creation

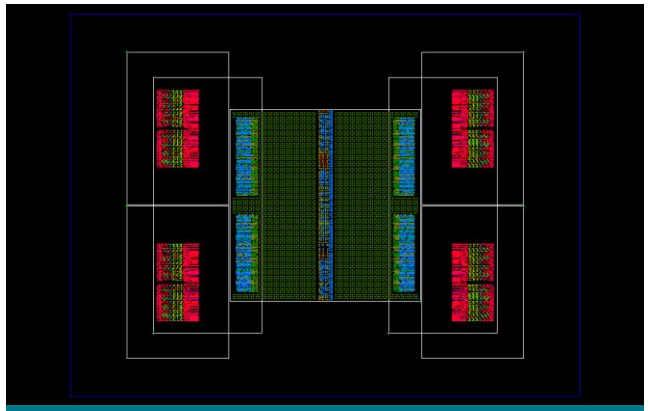
3Dblox
- 3dbx
- 3dbv

Collateral
- bmap
- upf
- lef
- tf
- v

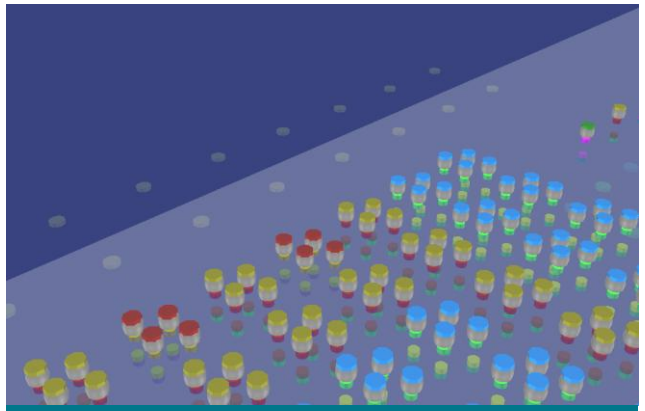
Read 3Dblox
- Create and stack chiplets
- Read technology files
- Import bumps and pins
- Build top-level logic

FTF
(Feedthrough Topology File)

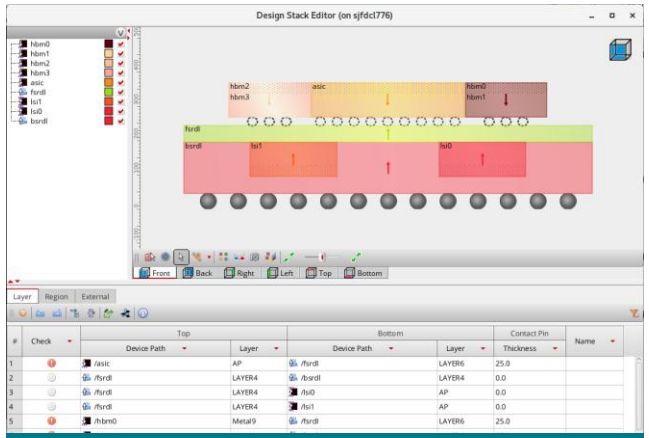
Feedthrough Insertion
- Create net groups
- Read FTF file
- Create feedthrough



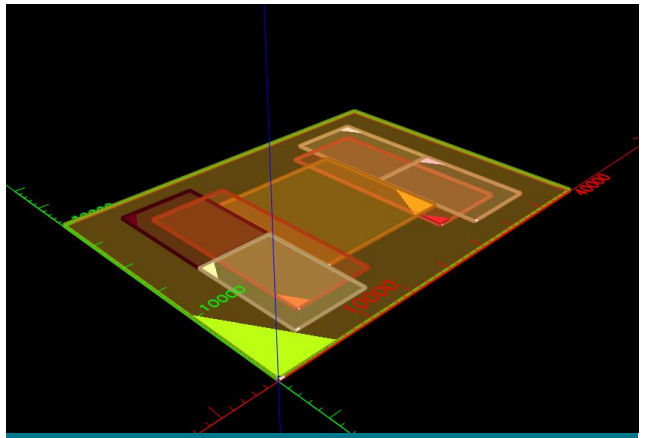
After planner::read_3dblox, pins and bumps from top dies are imported.



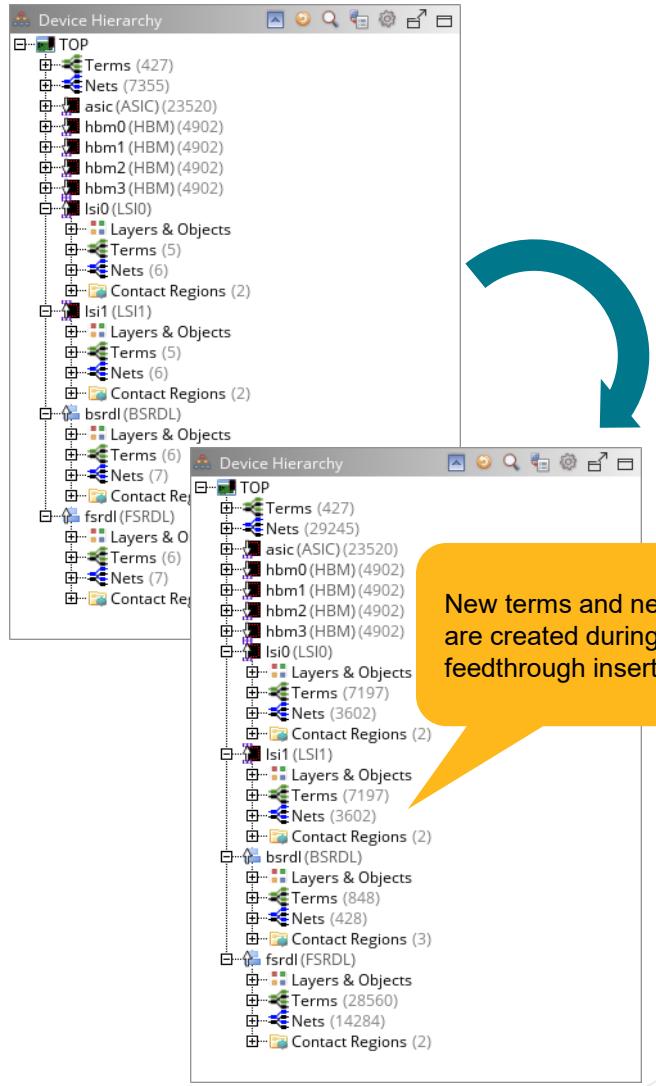
Design Detail 3D View



After planner::read_3dblox, design stack and contact layers are built.



Design Stack 3D View



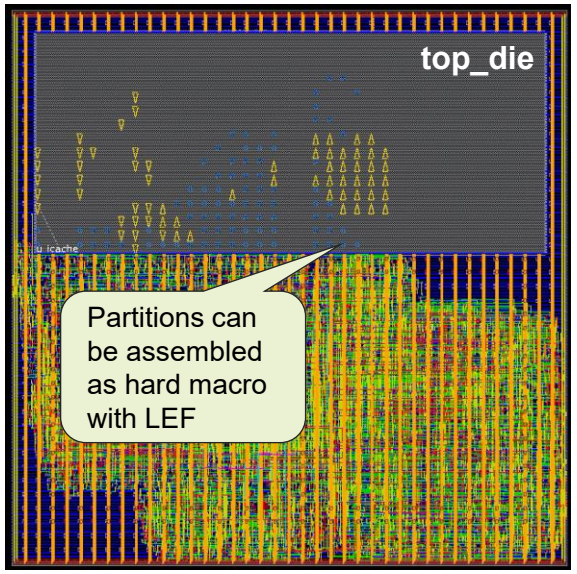
read_3dblox creates top-level logic and PG
create_feedthru generates full netlist

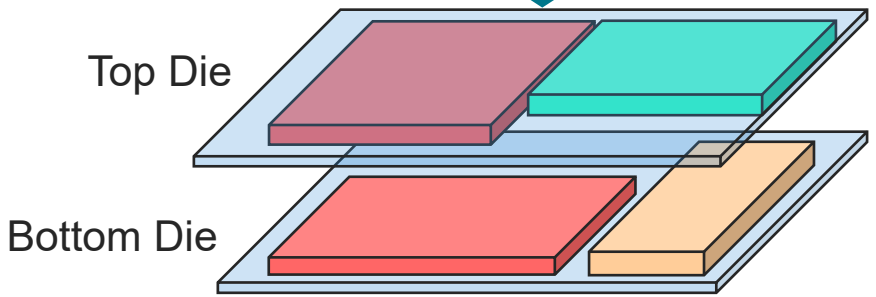
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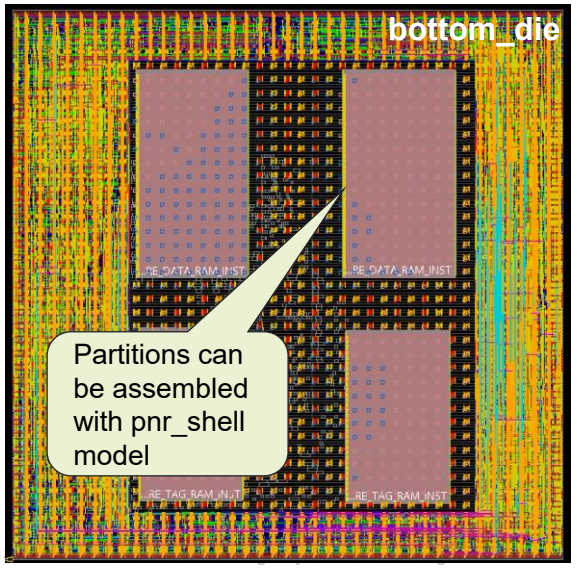
Top-Down and Bottom-Up Flows for 3D Stacking

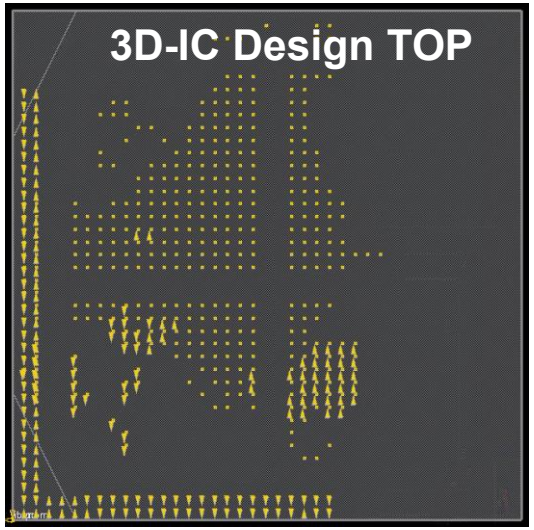
Bottom-up Flow

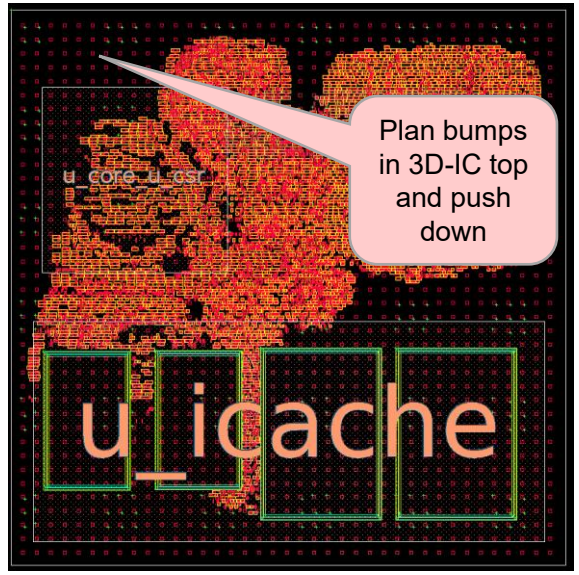
Assemble Innovus™ databases from each die to the 3D-IC top level.

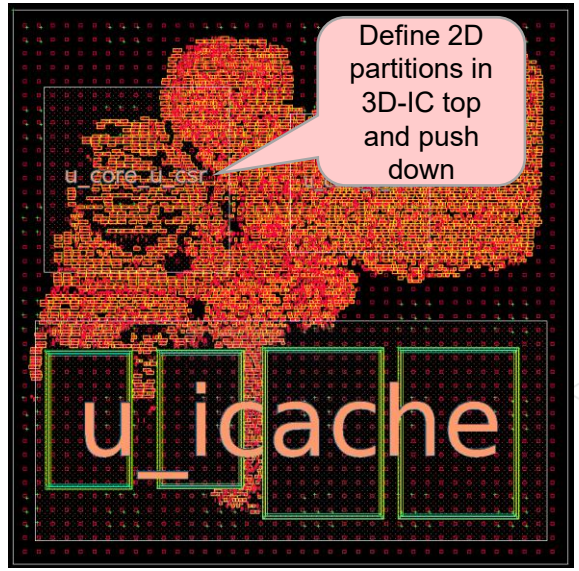












Top-Down Flow

Plan bumps and partitions in the 3D-IC top level and then push down

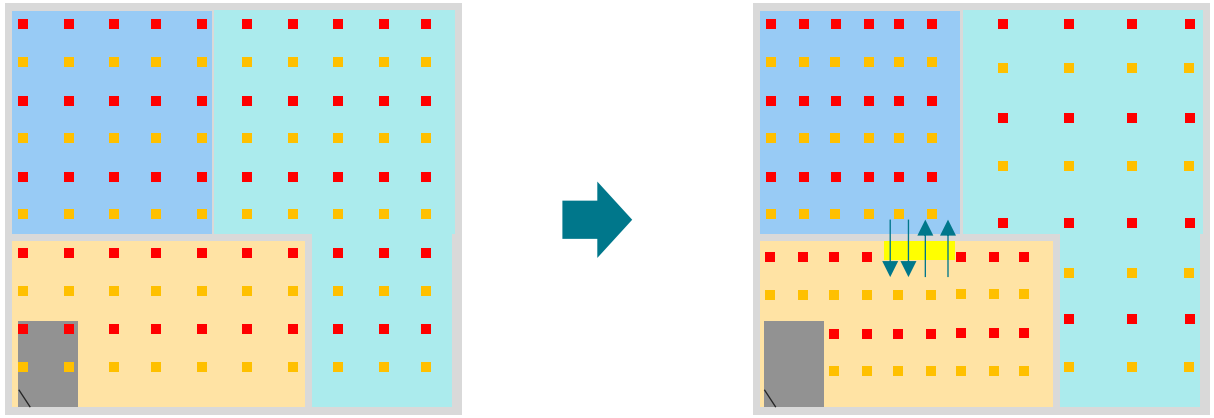


3D-IC TSV Distribution Optimization

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3D-IC TSV Distribution Optimization

- Provide user the ability to optimize TSV distribution. It helps users to:
  - Efficiently plan and allocate TSVs
  - Reduce IR-drop impact in stacked chiplets while satisfying the TSV density constraint
  - Perform detailed TSV placement in the floorplan and honor various kinds of floorplan constraints
- Supports
  - Optimize TSV distribution in different area considering the power region
  - Multiple type of PG bumps
  - Honor placement blockage and fixed macros
  - Avoid TSV assignment in high congestion area
  - Avoid partition pin assignment area
  - Can be used in both 2D and 3D design



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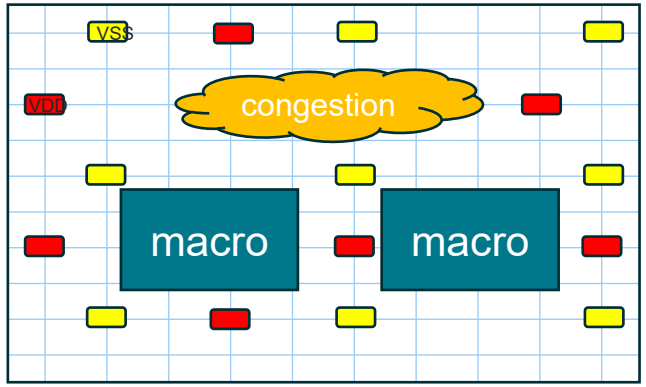
Power-Aware 3D-IC Optimization – PG TSV Placement Strategy

Fast and flexible PG TSV optimization using proto3d::opt_tsv_distribution

- **Key Enhancements**
  - **Fast Allocation**: Near real-time PG TSV placement accelerates design iteration
  - **Comprehensive Flow Support**: Supports both early floorplan (pre-PG) and post-PG optimization stages
  - **Placement Flexibility**: Supports both **inline** and **staggered** TSV patterns
  - **Distribution Strategy**: Even distribution across power domains using local current-based budgeting
  - **Congestion Avoidance**: Placement avoids congested zones, macros and placement blockage.
  - **3D PG TSV Optimization**: Enables power domain projection

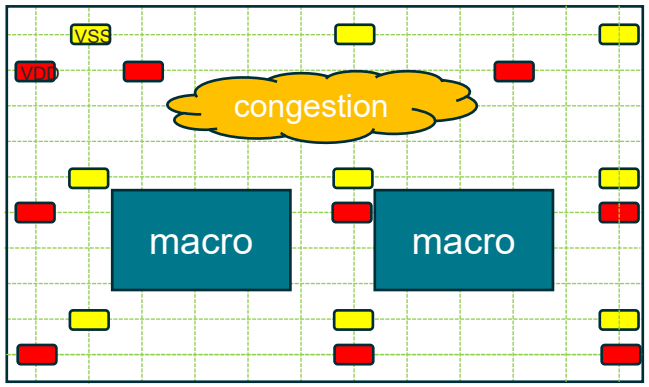
Early floorplan PG TSV Optimization

PG TSV Placement Based on Bump Grid




```
create_bump_grid -pitch 40 30 \
  -layers {Metal5 Metal6} -grid_name hb_bump_grid
```

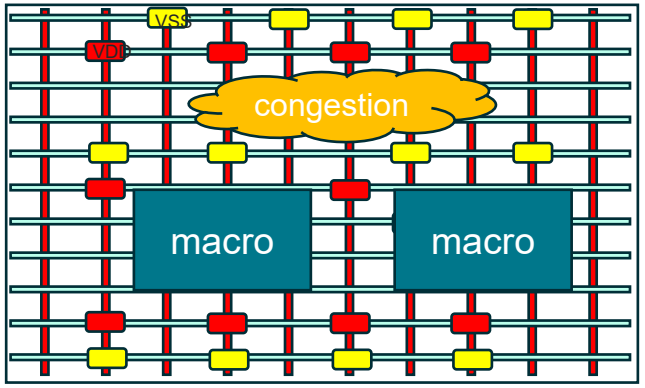
Placement Patterns and Direction




```
-pattern inline \
-pattern_direction horizontal \
```

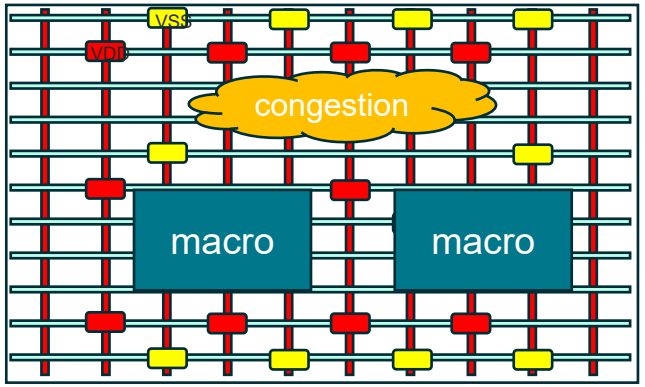
Post-PG PG TSV Optimization

Routing-Aware Considering Landing layer




```
-pg_routing_aware –landing_layers {M5 M6} \
-pattern inline \
-pattern_direction horizontal \
```

Placement Patterns and Direction




```
-pattern stagger \
-pattern_direction vertical \
```


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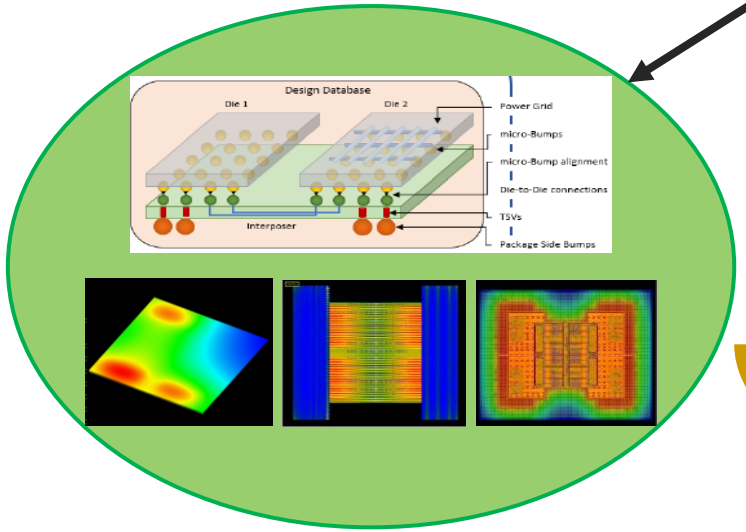
3D Chiplet Exploration and Prototyping

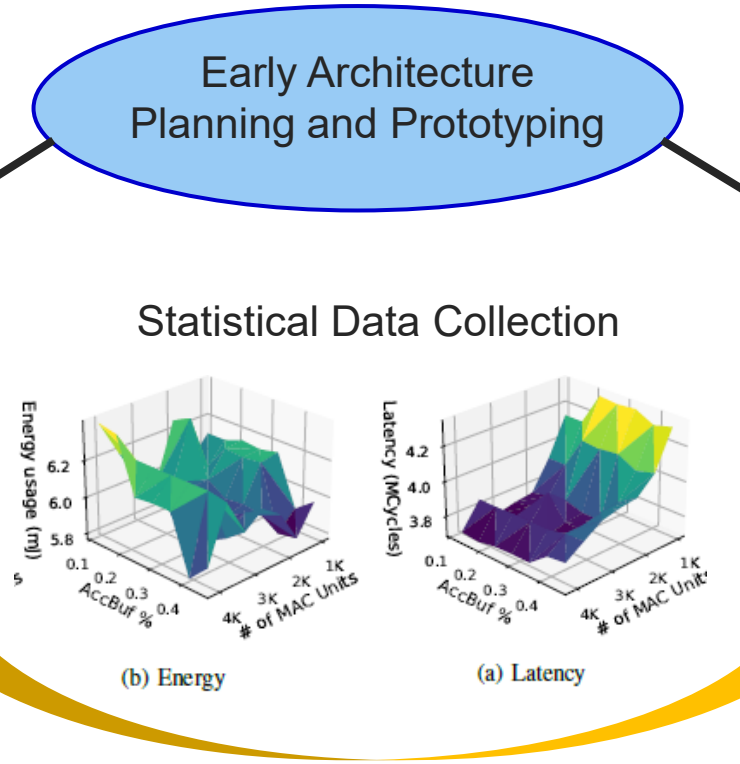
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Early Architecture Planning and Prototyping of 3D-IC Designs

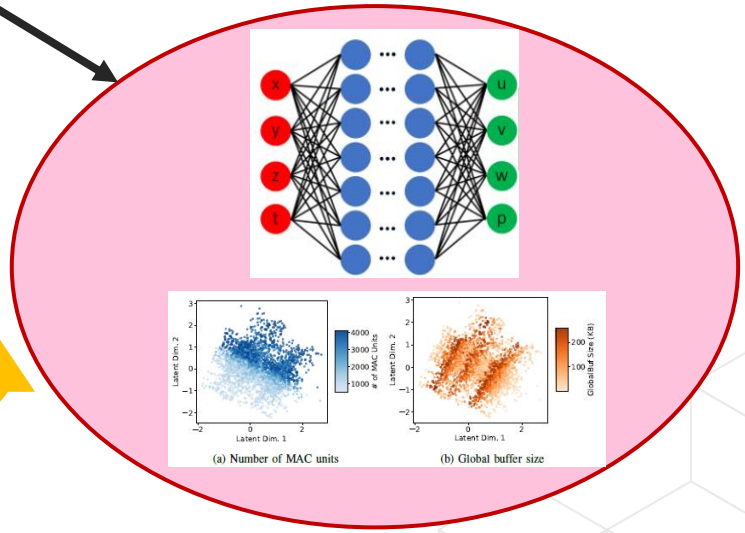
- Early-stage architecture planning and prototyping will aim at
  - Building the complex system from scratch or leveraging existing chiplets and IP to build comprehensive 3D-IC configurations
  - Enabling design feasibility and process technology evaluation, leading to more informed decisions about disaggregation choices and PPA tradeoffs
- Two strategic approaches may be taken as follows:

A simplified design or representative reduced netlist through fast implementation flow





A regression or ML model built upon given information/data

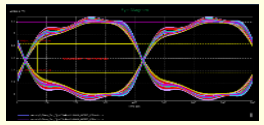


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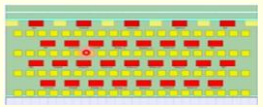
3D-IC Generic Exploration and Prototyping Framework

- A Cadence Cerebrus® based feature with ease-of-use built-in exploration functions
  - Driven by a user-friendly high-level YAML configuration file
  - Work with parameterized collaterals or template files easily

Interposer Routing Optimization

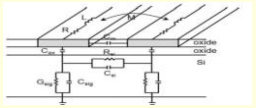


SI Simulation



Parameter Exploration

Cadence Cerebrus® Explorer

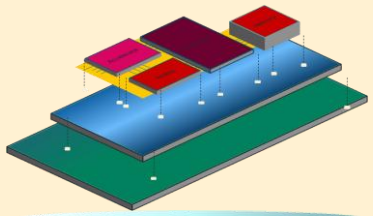


RLC Model Extraction



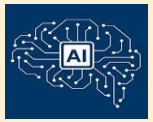
Interposer Routing

Partition, PDN+TSV, IR, and Thermal Exploration

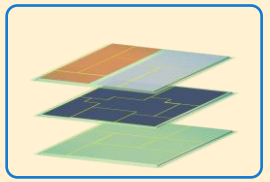


Integrity™ 3D-IC Platform

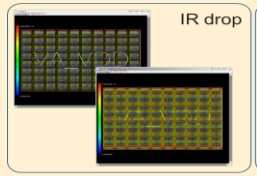
+



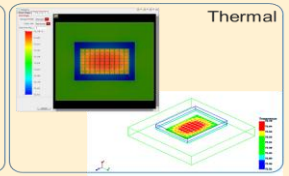
Cadence Cerebrus



Partition



PDN → IR Drop



Thermal

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Generic 3D-IC Exploration Flow with AI Approach
(Using Cadence Cerebrus® Explorer)

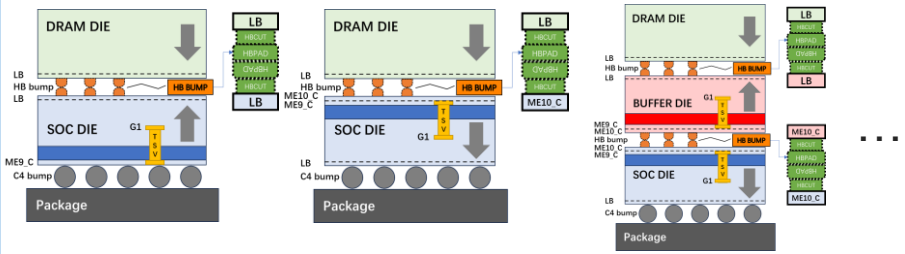
0% Netlist + without PDN plan — Phase0

- 3D-IC stack structure build up with 3DBlox
- Feasibility early IR and thermal analysis
- Bump feedback

0% Netlist + with PDN plan — Phase1

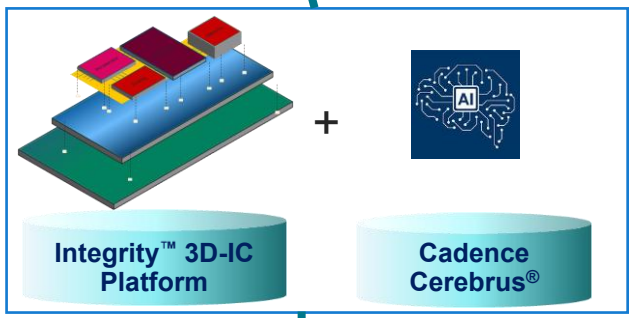
- Set up the design with 3DBlox
- Early IR and thermal analysis
- 3D-IC floorplan/partition
- PG bump planning
- System-level 3D PDN implementation (PSDL) and feedback

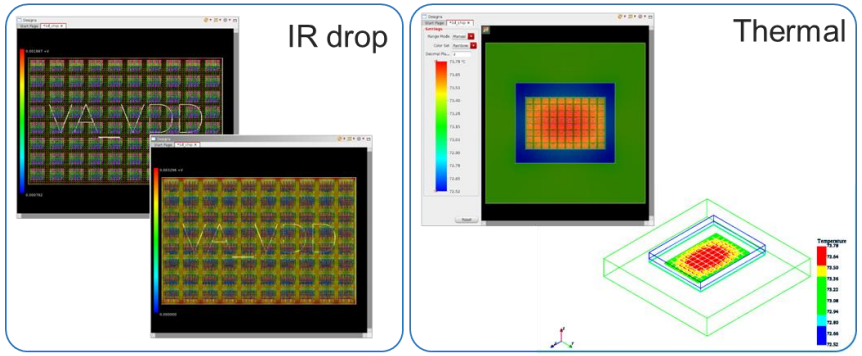
Different 3D-IC Prototyping Phase



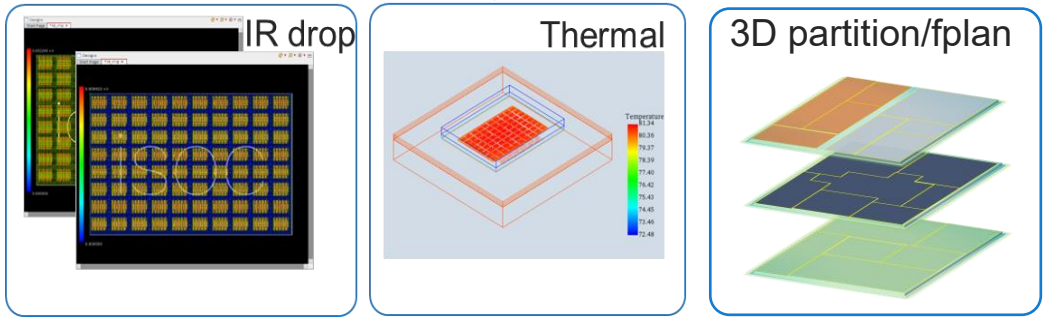
Different Package Options

3D-IC Exploration





Phase0: Feasibility IR and thermal



Phase1: Early IR, thermal, and 3D partition/fplan

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Integrity 3D-IC Platform: Early Exploration and Prototyping

0% netlist + without PDN plan

Phase0

- 3D-IC stack structure build up
- Feasibility early IR and thermal analysis
- Bump feedback

0% Netlist + with PDN plan

Phase1

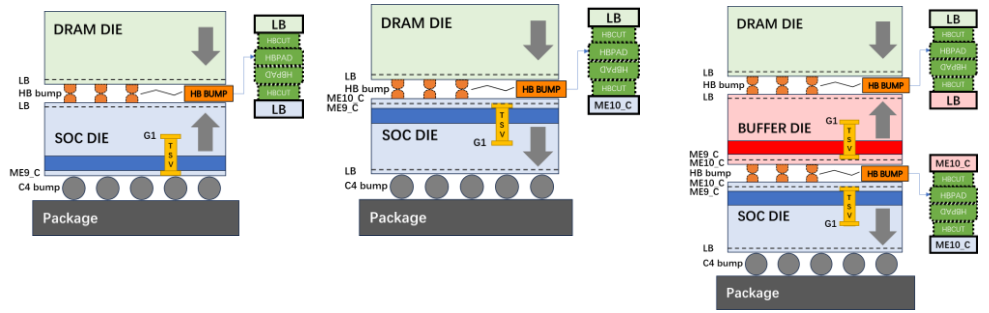
- Early IR and thermal analysis
- 3D-IC floorplan/partition
- PG bump planning
- System-level 3D PDN implementation (PSDL) and feedback

30% Netlist

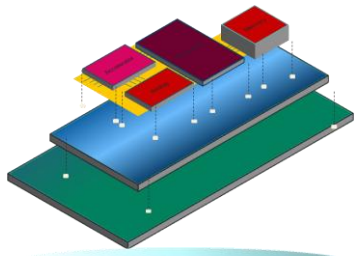
Phase2

- IR and thermal analysis
- 3D-IC placement and bump assignment
- Early STA preparation

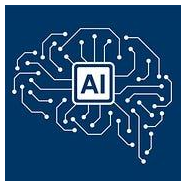
Different 3D-IC Prototyping Phase



Different Package Options



+



Integrity™ 3D-IC Platform

Cadence Cerebrus® Explorer

- Define various package options, 3D-IC design information, and other input files in main yaml file.
- Define primitives in primitive yaml

Multi-die Exploration and Optimization Using AI

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Technology Updates for 2.5D Style Designs

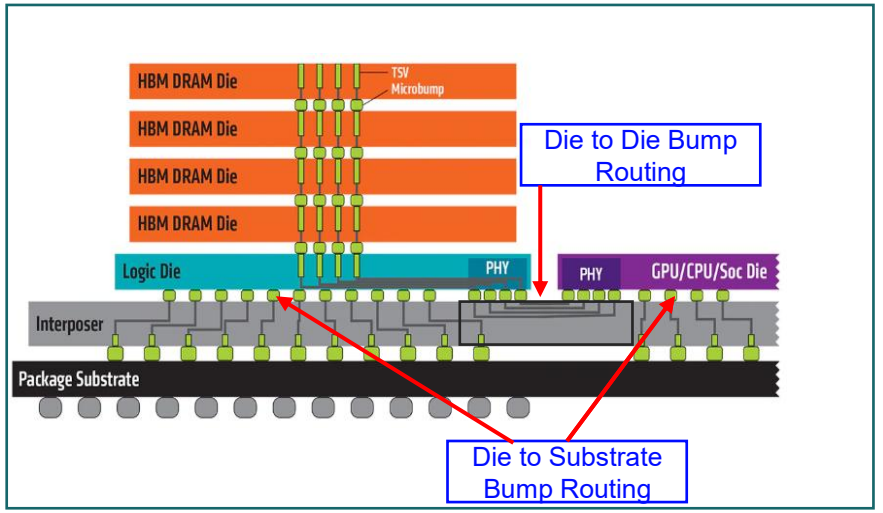
Si Interposer and Si-Bridge

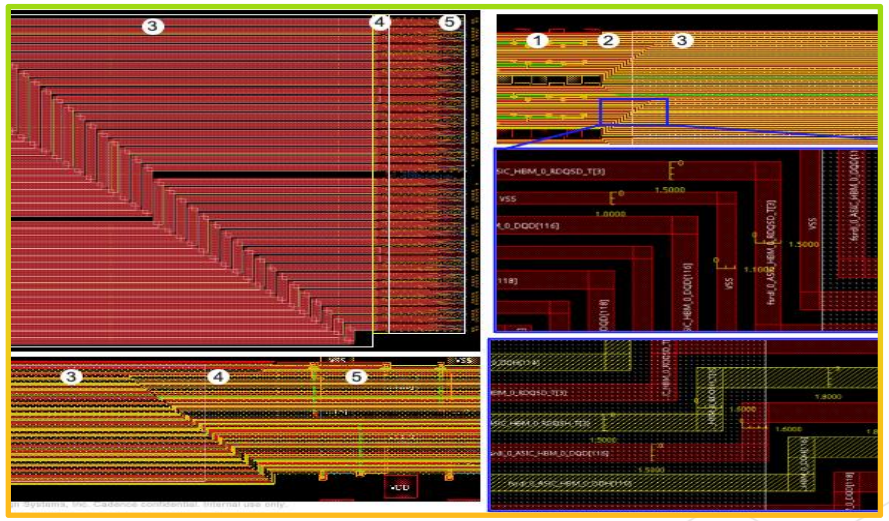
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Si Interposer Router Update

Updated for the latest chiplet routing requirements

- Requirements
  - Strict auto-routing requirements for new standards
  - Tighter routing constraints with shielding and smaller bump pitches
  - Many more custom patterns are necessary for maximizing the interposer route area
  - Need awareness of DRC rules and SI
- Cadence OmniRoute™ Advanced Interconnect Router
  - Designed from the ground up for chiplet routing on silicon interposer
  - Multi-threaded for millions of connections
  - Fast Runtime: Cadence OmniRoute has a very fast runtime, runs in a few seconds making it efficient for various routing tasks
  - Unified Routing Constraint Support: Supports unified routing constraints for Silicon Bridge Mixed Fabric designs
  - Support for 45-degree routing and shielding
  - Designed to have superior usability with HBM and UCIe™ recognition
  - Provides a seamless use-model for Nanoroute™ users

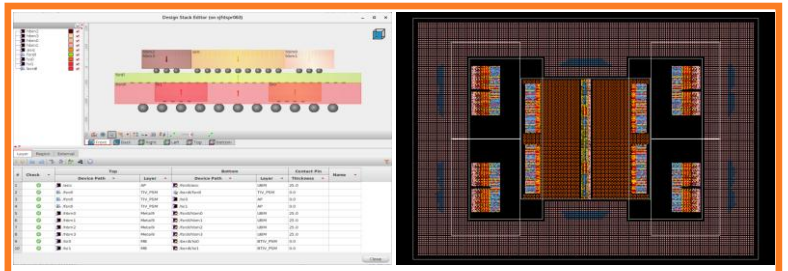




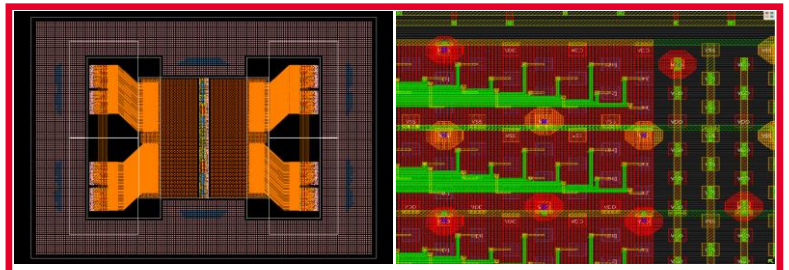
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Silicon Bridge End-to-End Solution

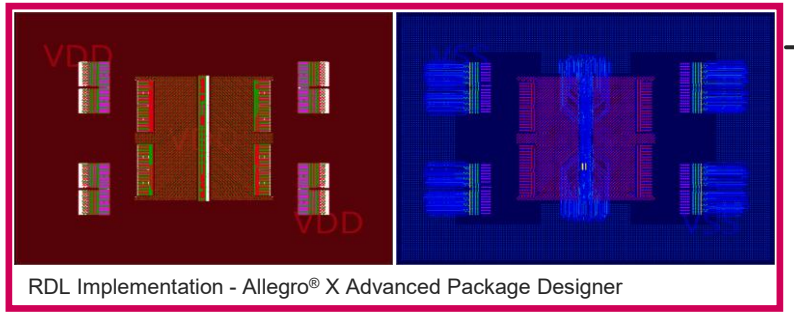
ASIC – HBM combination design with RDL and with Si bridges



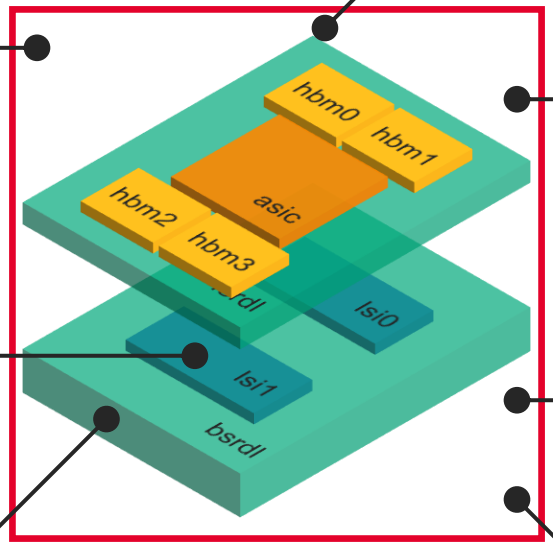
System Planning - Integrity™ System Planner



Silicon Routing - Innovus™ Implementation System

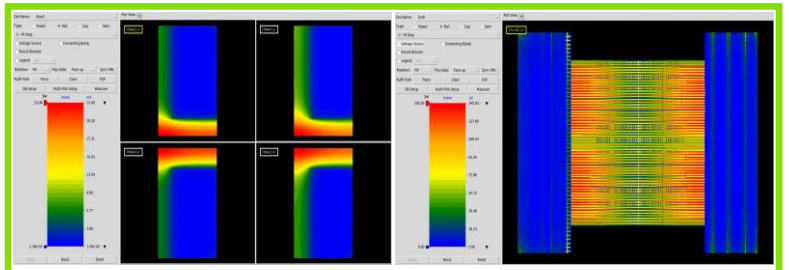


RDL Implementation - Allegro® X Advanced Package Designer

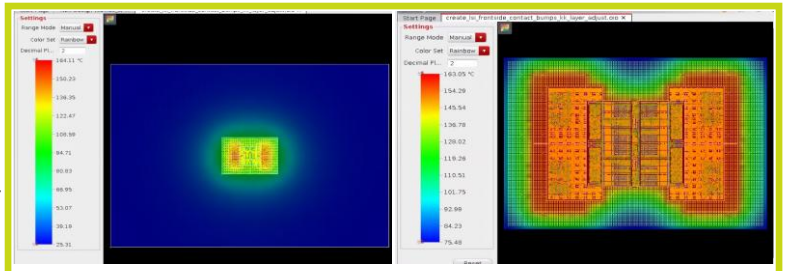


Integrity™ 3D-IC Platform

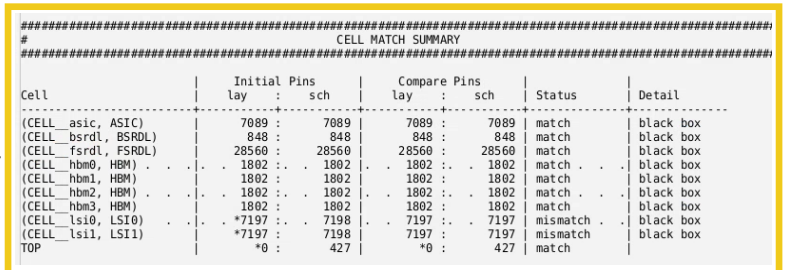




IR Drop Analysis - Voltus™ IC Power Integrity Solution



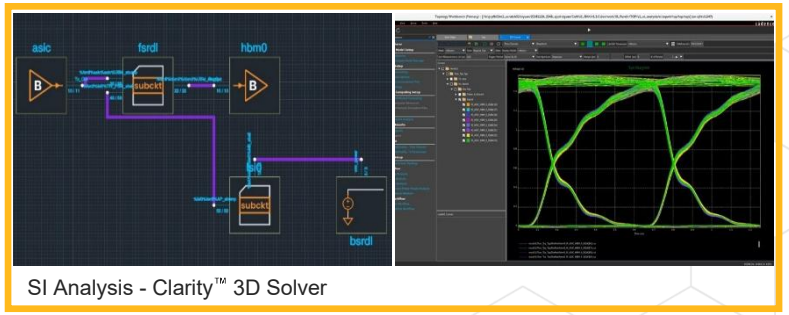
Thermal Analysis - Celsius™ Thermal Solver



```
#################################################################################################################
#                                                CELL MATCH SUMMARY
#################################################################################################################

                                |    Initial Pins     |    Compare Pins     |                |
Cell                            |    lay    :   sch   |    lay    :   sch   |  Status        |  Detail
--------------------------------+---------------------+---------------------+----------------+--------------
(CELL__asic, ASIC)              |       7089 :      7089 |       7089 :      7089 |  match         |  black box
(CELL__bsrdl, BSRDL)            |        848 :       848 |        848 :       848 |  match         |  black box
(CELL__fsrdl, FSRDL)            |      28560 :     28560 |      28560 :     28560 |  match         |  black box
(CELL__hbm0, HBM) . . .|. . 1802 :. . 1802 |. . 1802 :. . 1802 |  match . . .|  black box
(CELL__hbm1, HBM)               |       1802 :      1802 |       1802 :      1802 |  match         |  black box
(CELL__hbm2, HBM) . . .|. . 1802 :. . 1802 |. . 1802 :. . 1802 |  match . . .|  black box
(CELL__hbm3, HBM)               |       1802 :      1802 |       1802 :      1802 |  match         |  black box
(CELL__lsi0, LSI0)  . . .|. . *7197 :. . 7198 |. . 7197 :. . 7197 |  mismatch . . .|  black box
(CELL__lsi1, LSI1)              |       *7197 :      7198 |       7197 :      7197 |  mismatch      |  black box
TOP                             |         *0 :       427 |         *0 :       427 |  match         |
```

System LVS - Pegasus™ Verification System



SI Analysis - Clarity™ 3D Solver

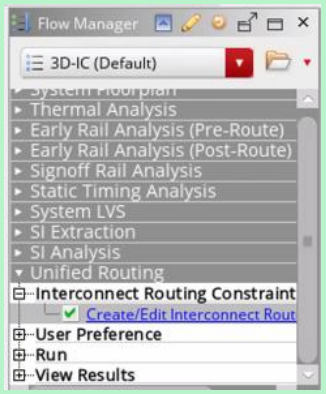
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Unified Routing Cockpit: Si+RDL Routing in Si Bridge Designs

Parallel routing implementation with unified routing cockpit (URC) in the Integrity 3D-IC Platform

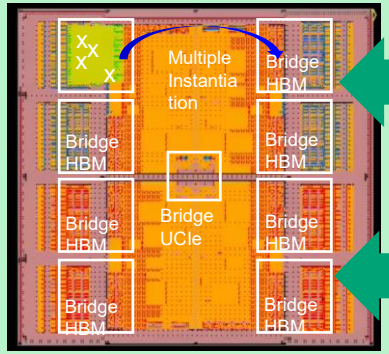
Unified Router Cockpit for Auto Routing

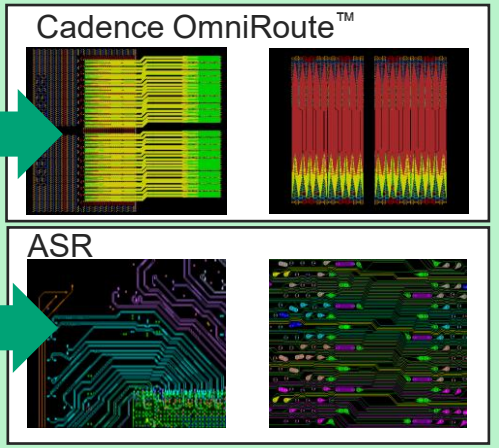
- Create/Edit Unified Constraint
- User Preference Setting
- Auto Routing
- Routed DB/DRC Back Annotation
- DRC/Connectivity Checking
- SI/PI Analysis Clarity™→TopWB
- Routing Constraint Update
- Save/Restore Database



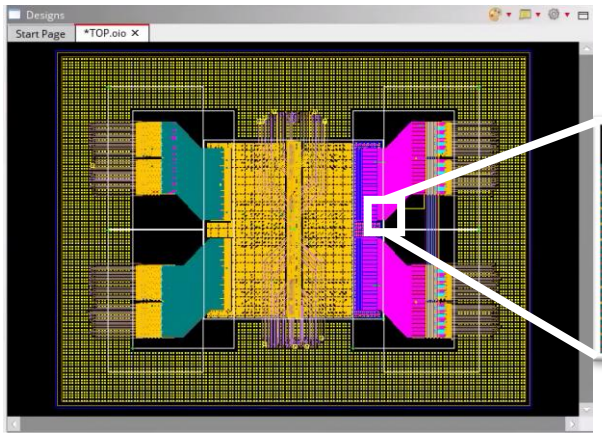
Unified Routing Cockpit

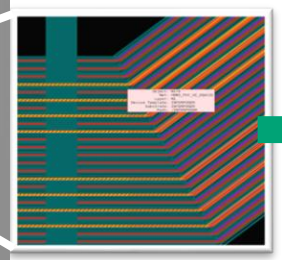
- Unified routing constraint
- Auto route for both die/RDL
- Parallel routing job control
- DRC browsing and LVS checking
- Fully scriptable and automatic solution

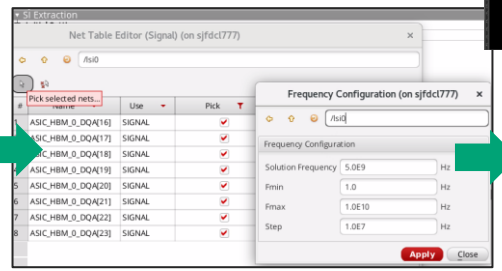


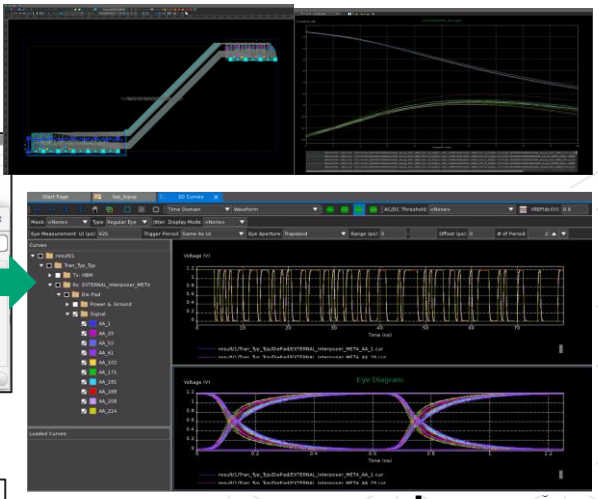


In-Design Seamless SI Analysis for Multi-Net Groups in Parallel







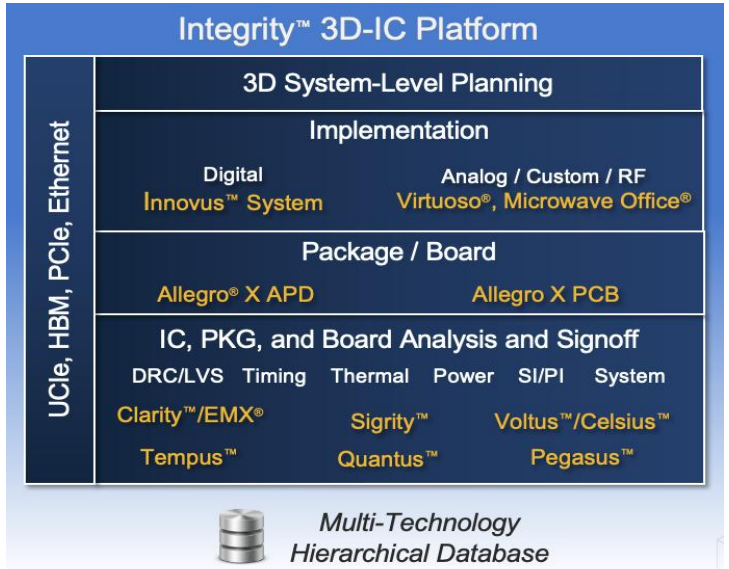


*Required Integrity 3D-IC build: **25.13-e009_1** or later version*

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Summary

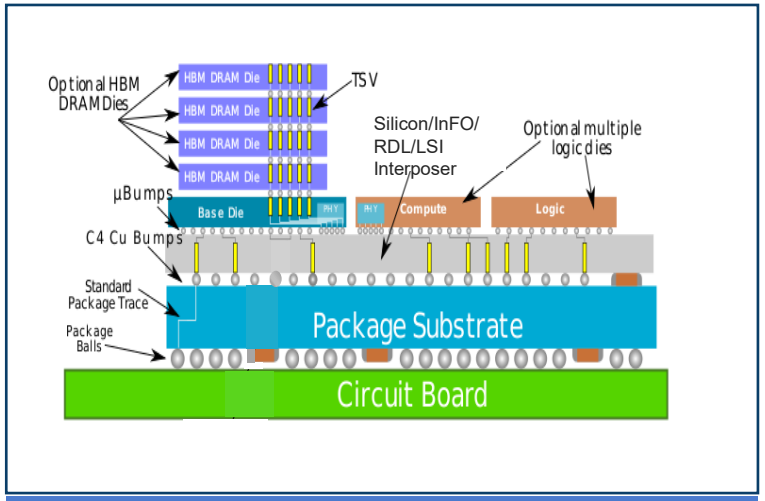
- Full chiplet stacking or side-by-side chiplet implementation flows available through Cadence's 3D-IC platform
- Multiple Testchip and real tapeouts with HBPs completed
- Basic and advanced features of 3Dblox™ available
- Detailed capabilities for chiplet planning, chiplet implementation, and signoff tapeout closure available
- Early exploration and prototyping flows available through Cadence Cerebrus® Explorer infrastructure

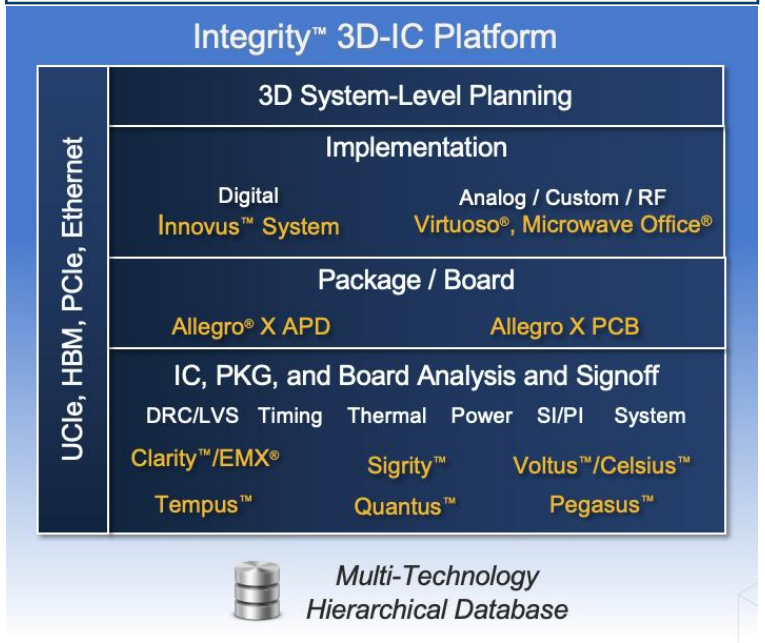


Flow Step	Activity	Details
3D-IC Design Entry	3Dblox file and stack definition	• Direct read of 3Dblox and iHDB creation. • 3Dblox Quality Checking
	Individual Die Design Entry	• Basic collateral for each individual die for iHDB population
3D-IC Prototyping	Die stacking with IR impacts	• Early Rail Analysis through iHDB setup
	Die stacking with Thermal impacts	• Early Thermal Analysis through iHDB setup
3D-IC Design Partitioning Bump Planning, & Implementation	Floorplan based Bump assignment and management	• Hierarchical top-Down/Bottom up Bump assignment flow
	BUMP/TSV planning (DIEs with different shrink-factors)	• TSV/BUMP projection flow considering Optical shrink
	SAI-driven early IR and thermal analysis	• SAI auto-generate a flatten netlist with region-based power data, enabling early IR drop and thermal analysis without full layout.
	BUMP/TSV Optimization flow	• TSV/BUMP analysis and congestion/timing/IR optimization
	BUMP alignment and connectivity check	• Basic Alignment checking & Alignment marker insertion flow.
	Seal Ring dummy HBI insertion	• Seal ring Tape-Out requirement from foundry
	3D-IC PG TSV planning and optimization	• Support w/o and w/ PG TSV optimization in inline and stagger patterns, preferred direction and landing layers for uniform IR drop.
	Backside RDL PG Routing	• Setup data to pkg tool for PG routing
3D-IC Signoff	3D System Level LVS and DRC checking	• Verilog vs Bump Layout connectivity checks.
	Inter-chiplet Antenna	• Integrity-Pegasus flow to check inter-chiplet antenna effect and generate the result directly.
	3D EM/IR	• Voltus 3D EM/IR flow – Pkg model Generation, VTM Generation • TSV cluster modeling
	3D RC extraction	• Quantus IDX flow for inter die SPF extraction
	3D STA	• Tempus 3D view reduction and STA flow for synchronous clocking designs

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Summary: End-to-End Flow for Complex Multi-Die Design





- System Partitioning and Planning for 2.5D or 3D Stacks
- 3D Structure Aware Implementation (TSV, backside metal)
- Thermal and Power (Early Analysis and Signoff)
- Integration (Analog and Package Co-design)
- Multi-Die Timing Signoff
- Multi-Die Physical Verification
- System/Package-Level Extraction SI/PI Analysis

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